

8.1

$$A = \begin{pmatrix} -1 & -2 & 0 \\ -1 & 3 & -1 \\ 2 & 3 & 1 \end{pmatrix}$$

Najdeme charakteristický polynom matice A .

$$\det \begin{pmatrix} -1-\lambda & -2 & 0 \\ -1 & 3-\lambda & -1 \\ 2 & 3 & 1-\lambda \end{pmatrix} = (-1-\lambda)(3-\lambda)(1-\lambda) + 4 - 3 - 3\lambda - 2 + 2\lambda =$$

$$= (-1-\lambda)(3-4\lambda+\lambda^2) - 1 - 1\lambda = -3 + 4\lambda - \lambda^2 - 3\lambda + 4\lambda^2 - \lambda^3 - 1 - 1\lambda =$$

$$= -\lambda^3 + 3\lambda^2 - 4 = (-\lambda-1)(\lambda^2-4\lambda+4) = (-\lambda-1)(\lambda-2)(\lambda-2)$$

Matice A má tedy vlastní čísla -1 algebraické násobnosti 1 a číslo 2 algebraické násobnosti 2.

Nalezneme vlastní vektory k těmto číslům.

$$M_1 = \text{Ker} \begin{pmatrix} 0 & -2 & 0 \\ -1 & 4 & -1 \\ 2 & 3 & 2 \end{pmatrix} = \text{Ker} \begin{pmatrix} 0 & -2 & 0 \\ -1 & 4 & -1 \\ 0 & 11 & 0 \end{pmatrix} = \text{Ker} \begin{pmatrix} 0 & -2 & 0 \\ -1 & 4 & -1 \\ 0 & 0 & 0 \end{pmatrix} = \text{Span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$M_2 = \text{Ker} \begin{pmatrix} -3 & -2 & 0 \\ -1 & 1 & -1 \\ 2 & 3 & -1 \end{pmatrix} = \text{Ker} \begin{pmatrix} -3 & -2 & 0 \\ -1 & 1 & -1 \\ 3 & 2 & 0 \end{pmatrix} = \text{Ker} \begin{pmatrix} -3 & -2 & 0 \\ -1 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} =$$

$$= \text{Ker} \begin{pmatrix} -1 & 1 & -1 \\ 0 & -5 & 3 \\ 0 & 0 & 0 \end{pmatrix} = \text{Span} \left\{ \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} \right\}$$

Protože 2 má geometrickou násobnost 1 dopocítáme zobecněný vlastní vektor:

$$\left(\begin{array}{ccc|c} -3 & -2 & 0 & -2 \\ -1 & 1 & -1 & 3 \\ 2 & 3 & -1 & 5 \end{array} \right) = \left(\begin{array}{ccc|c} -1 & 1 & -1 & 3 \\ -3 & -2 & 0 & -2 \\ 2 & 3 & -1 & 5 \end{array} \right) = \left(\begin{array}{ccc|c} -1 & 1 & -1 & 3 \\ 0 & -5 & 3 & -11 \\ 0 & 5 & -3 & 11 \end{array} \right) =$$

$$= \left(\begin{array}{ccc|c} -1 & 1 & -1 & 3 \\ 0 & -5 & 3 & -11 \\ 0 & 0 & 0 & 0 \end{array} \right) = \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}$$

Tedy báze $B = \text{span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix} \right\}$

a h ní je Jordanova matice $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$

Tedy $U_1 = \text{span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$, $A \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ tedy se zosazí do U_1

Báze $B_{U_1} = \text{span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$ a

$$\left[f_A|_{U_1} \right]_{B_{U_1}}^{B_{U_1}} = (-1)$$

A pro $U_2 = \text{span} \left\{ \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix} \right\}$ platí

$$A \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} = \begin{pmatrix} -4 \\ 6 \\ 10 \end{pmatrix} \quad \text{a} \quad A \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} -2 \\ 5 \\ 1 \end{pmatrix}$$

Tedy $f(u) = 2u$ a $f(v) = u + 2v$

A pro $B_{U_2} = \text{span} \left\{ \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix} \right\}$ a

$$\left[f_A|_{U_2} \right]_{B_{U_2}}^{B_{U_2}} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$